Technical Document #732: Retrieval Augmented Generation - Comprehensive Overvi

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Hybrid search combines keyword-based and semantic search. It leverages the strengths of both approaches for better retrieval quality.

Vector databases store dense embeddings for semantic search. Pinecone, Weaviate, and ChromaDB are popular vector database solutions.

Chunking splits documents into smaller segments for embedding. Optimal chunk size balances context retention with retrieval precision.

Retrieval-Augmented Generation (RAG) combines information retrieval with text generation. It grounds LLM responses in factual, retrieved documents.

Semantic search finds documents based on meaning rather than exact keywords. It uses embedding similarity to match queries with relevant content.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Key Takeaways:

- Query decomposition breaks complex queries into simpler sub-queries.
- Retrieval-Augmented Generation (RAG) combines information retrieval with text generation.
- Re-ranking models score retrieved documents for relevance to the query.